Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

FIIT-XXXX-XXXXX

Bc. Richard Čerňanský

Deep Learning Methods for Perception and Prediction in Autonomous Systems

Master's Thesis

Thesis Supervisor: Ing. Matej Halinkovič, PhD.

May 20xx

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

FIIT-XXXX-XXXXX

Bc. Richard Čerňanský

Deep Learning Methods for Perception and Prediction in Autonomous Systems

Master's Thesis

Study Programme:	Intelligent Software Systems
Study Field:	Computer Science
Training Workplace:	Bratislava, Slovakia
Thesis Supervisor:	Ing. Matej Halinkovič, PhD.

May 20xx

Čestne vyhlasujem, že som túto prácu vypracoval samostatne, na základe konzultácií a s použitím uvedenej literatúry.

I declare in my honor that I have prepared this work independently, on the basis of consultations and using the mentioned literature.

In Bratislava, May 20xx

Bc. Richard Čerňanský

Annotation

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies
Study Programme: Intelligent Software Systems
Study Field: Computer Science

Author: Bc. Richard Čerňanský
Master's Thesis: Deep Learning Methods for Perception and Prediction in Autonomous Systems
Thesis Supervisor: Ing. Matej Halinkovič, PhD.
May 20xx

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Keywords: 3-5 keywords

Anotácia

Slovenská technická univerzita v Bratislave
Fakulta informatiky a informačných technológií
Študijný program: Inteligentné softvérové systémy
Študijný odbor: Informatika

Autor: Bc. Richard Čerňanský
Diplomová práca: Deep Learning Methods for Perception and Prediction in Autonomous Systems
Vedúci diplomového projektu: Ing. Matej Halinkovič, PhD.
Máj 20xx

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Oblasť problematiky: 3-5 kľúčových slov

BP2, DP3: sem vložte finalne a podpisane zadanie: PDF z AISu

BP1, DP1, DP2: sem vložte navrh zadania: PDF z YonBanu

Acknowledgements / Poďakovanie

The acknowledgments and the people to thank go here, don't forget to include your project advisor. . .

I also thank my family and friends who supported me during the writing of this thesis.

Contents

Chapter 1

Introduction

In recent years, deep learning and computer vision methods have made significant advancements in the analysis of dynamic scenes and their applications in autonomous systems. The development of fields such as autonomous vehicles and robotics depends on solving tasks like object detection, object tracking, segmentation, and trajectory prediction. Specifically, autonomous vehicles (AVs) revolutionize transportantion by saving drivers' time, improving safety and possibly mitigating congestion on the roads. Current autonomous driving systems consist of multiple sequential tasks such as perception, tracking, prediction, planning and control [1]. Perception task is responsible for detecting and understanding the objects within the environment including their categorization, location and size estimation. The tracking objective uses the detections from perception to find the relationships between objects across the frames in scene. The prediction modules then process this tracking information in order to forecast future positions. The predicted positions enter the planning stage that generates the most feasible and optimal path in terms of safety, energy consuption and following traffic rules. Last but not least, controlling task then prepares actuator commands which execute the planned motion.

1.1 Motivation

Real traffic is highly dynamic and unpredictable: other drivers may behave erratically, and sensors provide noisy or incomplete data. Diverse maneuvers, complex interactions, and inherent errors in sensory information make prediction challenging, yet solving these challenges is essential for having trustworthy AVs. In addition, autonomous systems operate under strict constraints on latency and reliability [2]. Predictions must be computed in real time on embedded hardware and remain accurate over long horizons to be useful for driving in real-life scenarios. Industry recognizes the need for fast execution, uncertainty handling, and generalization in prediction models to ensure robust operation in varied conditions [3, 4, 5]. Reliable trajectory prediction is therefore critical for safe autonomous driving and motivates intensive research into methods that can forecast the multi-agent future with practical efficiency. For these reasons, this thesis focuses on trajectory prediction, exploring how current state-of-the-art models could be modified or extended to achieve better accuracy and efficiency.

1.2 Problem Formulation

trajectory prediction task definition, mathematically

1.2.1 Data Fusion

Combine cameras, LiDAR or radar, and HD maps into one scene view. Compare early, BEV or attention-based intermediate fusion, and late fusion. Prevent overload or overspecialization through learned alignment and robust training [6].

1.2.2 Dynamic Environments

Traffic is nonstationary and many futures are plausible. Aleatoric uncertainty means the same history can lead to different outcomes. Predict a calibrated set of socially compliant trajectories so planning can weigh alternatives in real time [6].

1.2.3 Interaction Modeling

Agent motions are coupled and constrained by lanes and rules. Use structured representations such as graphs and attention to capture agent to agent and agent to map relations. Poor interaction modeling leads to infeasible predictions [2].

1.2.4 Real-World Constraints

The system must run within an onboard real-time loop and scale to dense traffic. Try to balance model capacity with memory and compute efficiency to stay deployable, not only accurate [4, 2].

1.3 Thesis Outline

Chapter 2

Overview of Trajectory Prediction Methods

concrete header for the given thesis's theme

2.1 Background

Don't write too long about well-known facts in books and text-books.

2.2 Related Work (to your work)

This part is important in determining the objectives of the Thesis as we will state in Section 2.3 and in Chapter 3.

2.3 Summary and Starting Points

Summary of Chapter 2 ... several paragraphs... 1/2 to 1 page

Based on the study presented in Chapter 2, the following gaps / needs are identified with potential to improve:

- question / gap / need
- ...
- question / gap / need

To answer these questions, the following starting points are proposed.

- Starting point 1:
- Starting point 2:

- Starting point 3:

Priebežná správa BP1/DP1 musí obsahovať:

1. Prehľad podobných riešení alebo produktov, s ktorými sa študent oboznámil pri štúdiu danej problematiky.
2. Špecifikáciu požiadaviek na vytvárané riešenie alebo produkt (Chapter 3).
3. Návrh
4. Vlastne zhodnotenie práce za príslušné obdobie (Chapter C - Section C.1).

Chapter 3

Objectives and Methodology

3.1 Objectives

Vybrané ciele pre navrhnuté riešenie alebo špecifikácia požiadaviek (funkčné a nefunkčné požiadavky) na naimplementovaný softvérový produkt.

- povinne pre DP a DiZP, ale aj BP môže
- max 2 pages for the whole chapter.

As presented in Chapter 2, the area of... is wide with many interesting unsolved problems. Therefore, the goals of this thesis are focused on selected objectives (maximum 3 objective) to... as follows:

- Objective 1:
- Objective 2:
- Objective 3:

3.2 Methodology

The following steps have been chosen for the design and development of ...

- Step 1:
- Step 2:
- ...
- Step n:

These steps are commitments to each other in the context of ... of the work.

Chapter 4

Design / Methods Description

Concrete header for the proposed design of the thesis's solution

This chapter can contain:

- Specification based on the Sections 2.3
- Design
- Technologies (can also be/moved in/to Chapter 5 Implementation)

Description how the design works (as much as possible from the below list)

- We need to separate our design from our concrete implementation
- Architecture, Block scheme, Event diagram, Sequence diagram, Algorithms, ...
- Text description of each blocks and their connections and/or dependencies
- Time complexity and memory complexity estimation for each execution step and the whole design. Communication complexity of the design (if any).

Figure 4.1 is about ... description of each blocks

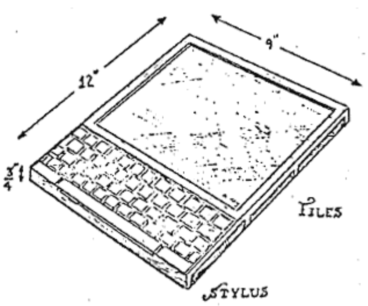

Figure 4.1: Architecture/Workflow

Chapter 5

Implementation

Concrete header for the implementation of the proposed design

The concrete realization of our proposed design

Concrete tools, software, solutions based on concrete dataset

Pseudocode of Algorithms, or Code snippets

$$\text{Bounce Rate} = \frac{\text{Single page sessions}}{\text{Total sessions}} \tag{5.1}$$

Chapter 6

Experiments and Evaluation

6.1 Setup

hardware,
software, libraries, versioning,
virtual environment description and setting,
datasets, NDA,
etc.

Machine	Remote	Google Colab	Kaggle	FIIT Cloud	Local
Python	3.9.19	3.10.12	-	-	3.10.12
PyTorch	-	-	-	-	-
TensorFlow	2.10	2.15	-	-	2.14
GPU (NVIDIA)	RTX xx	Tesla T4 15 GB	Tesla P100 16GB	-	-
CUDA	12.2	12.2	-	-	-
cuDNN	8.1.0	N/A	-	-	-
CUDA Toolkit	11.2.2	N/A	-	-	-
RAM	32/64 GB	51 GB	30 GB	32 GB	16 GB
CPU	8x cores Intel i7	-	Xeon 4x cores	4x cores	-
Storage	SSD	Google Driver	73 GB	128 GB	HDD

Table 6.1: Hardware, software and versioning used for realization

6.2 Case Study - Experiment 01

6.3 Case Study - Experiment 02

6.4 Case Study - Experiment 03

6.5 Summary

Main *key findings* of the work based on the Experiment results

- add ...
- ...

Discussion about various considerations based on the experiment results (if needed).

Regarding *limitations* of the work ...

- add ...
- ...

Chapter 7

Conclusion

7.1 Summary

Summary of the whole work including SOTA, design, implementations, experiments, evaluations, key fingdings ... 1 page

7.2 Future Work

1/2 page

Resumé

Chapter je bez cislovania, pouziva sa len **boldovanie textu** na zyraznenie useku textu. Bez Figure, Table, atd. ak treba tak odvolava sa na originalne Figure a Table.

10% v slovenčine pre tých, ktorí majú prácu v EN: preklad

EN	SK
Introduction	primerane strucne
SOTA	primerane strucne
SOTA - cast Summary and Starting Points	cele
Objective and Methodology	cele
Design, Implementation	primerane strucne
Experiments and Evaluation	primerane strucne
Experiments and Evaluation - cast Summary	cele
Conclusion and Future Work	cele

Table 7.1: Resumé

List of Abbreviations

AVs autonomous vehicles

List of Figures

List of Tables

References

[1] Li Chen et al. "End-to-end autonomous driving: Challenges and frontiers". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.12 (2024), pp. 10164–10183. DOI: `10.1109/TPAMI.2024.3435937`.

[2] Yanjun Huang et al. "A Survey on Trajectory-Prediction Methods for Autonomous Driving". In: *IEEE Transactions on Intelligent Vehicles* 7.3 (2022), pp. 652–674. DOI: `10.1109/TIV.2022.3167103`.

[3] Scott Ettinger et al. "The Waymo Open Motion Dataset: A Large-Scale Benchmark for Motion Forecasting". In: *Proceedings of the Conference on Robot Learning (CoRL)*. 2021.

[4] Nay and et al. "Wayformer: Motion Forecasting via Simple and Efficient Attention Networks". In: *arXiv preprint arXiv:2207.05844* (2022). URL: `https://arxiv.org/abs/2207.05844`.

[5] *Road vehicles — Safety of the intended functionality (SOTIF)*. International Organization for Standardization, 2022.

[6] Nadya Abdel Madjid et al. "Trajectory Prediction for Autonomous Driving: Progress, Limitations, and Future Directions". In: *arXiv preprint arXiv:2503.03262* (2025). URL: `https://arxiv.org/abs/2503.03262`.

Appendix A

Description of Digital Submission

Thesis Evidence Number:
FIIT-XXXX-XXXXX

Obsah elektronického média je väčší ako `1GB` a preto je odovzdaný vedúcemu záverečnej práce (doc. Ing. Giang Nguyen Thu, PhD.).

Name of submitted archive: `DP_MenoPriezvisko.zip`

The digital part included in the thesis contains the following files:

- Ubuntu OS: run in command line `$ tree` to get a directory structure as below.
- Inspiration from *https://github.com/FIIT-ISA/cookiecutter-data-science*

```
ROOT // Root directory.
│
├── flatbuffers // This flatbuffers director
└── scheme
```

```
├── LICENSE
├── Makefile           <- Makefile with commands like `make data` or `make train`
├── README.md          <- The top-level README for developers using this project.
├── data
│   ├── external       <- Data from third party sources.
│   ├── interim        <- Intermediate data that has been transformed.
│   ├── processed      <- The final, canonical data sets for modeling.
│   └── raw            <- The original, immutable data dump.
│
├── docs               <- A default Sphinx project; see sphinx-doc.org for details
│
├── models             <- Trained and serialized models, model predictions, or model summaries
│
├── notebooks          <- Jupyter notebooks. Naming convention is a number (for ordering),
│                         the creator's initials, and a short `-` delimited description, e.g.
│                         `1.0-jqp-initial-data-exploration`.
│
├── references         <- Data dictionaries, manuals, and all other explanatory materials.
│
├── reports            <- Generated analysis as HTML, PDF, LaTeX, etc.
│   └── figures        <- Generated graphics and figures to be used in reporting
│
├── requirements.txt   <- The requirements file for reproducing the analysis environment, e.g.
│                         generated with `pip freeze > requirements.txt`
│
├── setup.py           <- makes project pip installable (pip install -e .) so src can be impor
├── src                <- Source code for use in this project.
│   ├── __init__.py    <- Makes src a Python module
│   │
│   ├── data           <- Scripts to download or generate data
│   │   └── make_dataset.py
│   │
│   ├── features       <- Scripts to turn raw data into features for modeling
│   │   └── build_features.py
│   │
│   ├── models         <- Scripts to train models and then use trained models to make
│   │   │                 predictions
│   │   ├── predict_model.py
│   │   └── train_model.py
│   │
│   └── visualization  <- Scripts to create exploratory and results oriented visualizations
```

Figure A.1: The structure of the ZIP file

Appendix B

Technical Documentation

B.1 Installation Manual

1. Clone the project from the Git repository (URL) OR unzip the ZIP file ...
2. Run *pip install -r requirements.txt* in the root folder of the project
3. Run *python -m spacy download en* to have the data corpus ...

B.2 User Manual / Používateľská príručka

1. Spustenie programu funguje pomocou hlavného Python súboru *main.py* príkazom *python main.py*.
2. Po spustení programu sa zobrazí grafické rozhranie programu v ktorom sa dá hneď na hlavnej obrazovke spustiť sumarizácia pre jeden vstupný súbor.
3. Tlačidlami v hornej časti obrazovky sa dá presúvať medzi funkcionalitami programu ako je viac súborová sumarizácia alebo zobrazovanie článkov. Na týchto obrazovkách sa tlačidlom *Back to main menu* dá vrátiť späť na hlavnú obrazovku.

Screenshot(s) používateľského grafického rozhrania (GUI) ako obrázok (Figure)

(Another variant without GUI:) The main objective of this work is to ... Several case studies were presented that examine the capabilities of ... to show the feasibility and achievability of ... Users in our work are considered to work directly with the code as described in Section B.1.

B.3 Developer Manual

if needed then add ...

Appendix C

Work Schedule

C.1 Work Schedule for BP1/DP1

Week	Plan
1. - 2.	...
3. - 4.	...
5. - 6.	...
7. - 8.	...
9. - 10.	...
11. - 12.	...

Some explanation text... (if needed) ... a few sentences

Self-Evaluation of Work Schedule for BP1/DP1: Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The BP/DP 1 work plan is considered satisfied and successful.

C.2 Work Schedule for BP2/DP2

Week	Plan
1. - 2.	...
3. - 4.	...
5. - 6.	...
7. - 8.	...
9. - 10.	...
11. - 12.	...

Some explanation text... (if needed) ... a few sentences

Self-Evaluation of Work Schedule for BP2/DP2: Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The BP/DP 2 work plan is considered satisfied and successful.

C.3 Work Schedule for DP3

Week	Plan
1. - 2.	...
3. - 4.	...
5. - 6.	...
7. - 8.	...
9. - 10.	...
11. - 12.	...

Some explanation text... (if needed) ... a few sentences

Self-Evaluation of Work Schedule for DP3: Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The DP 3 work plan is considered satisfied and successful.